

Laser powder bed fusion of a Nb-based refractory alloy: Microstructure and tensile properties

Jianan Chen^a, Wangwang Ding^a, Qiying Tao^a, Chuanzhen Ma^b, Cong Zhang^b, Gang Chen^{a,c,*}, Mingli Qin^{a,c}, Xuanhui Qu^{a,c,**}

^a Beijing Advanced Innovation Center for Materials Genome Engineering, Institute for Advanced Materials and Technology, University of Science and Technology Beijing, Beijing, 100083, PR China

^b Collaborative Innovation Center of Steel Technology, University of Science and Technology Beijing, Beijing, 100083, PR China

^c Beijing Laboratory of Metallic Materials and Processing for Modern Transportation, Beijing, 100083, PR China

ARTICLE INFO

Keywords:

Additive manufacturing
Powder modification
Niobium alloy
Microstructure
Mechanical properties

ABSTRACT

In this study, Nb521 (Nb–5W–2Mo–1Zr) parts were successfully fabricated by laser powder bed fusion (LPBF) using quasi-spherical hydride-dehydride (HDH) powders modified by jet milling. The relative density of the LPBF made (LPBFed) Nb521 parts is $98.8 \pm 0.2\%$, which were subsequently subjected to hot isostatic pressing (HIP) to achieve almost full densification. Nanoscale ZrO_2 precipitates were found in the LPBFed parts and those after HIP (HIPed), showing semi-coherent with the matrix. The precipitation behavior is strongly related to the melt pool solidification process during LPBF in terms of thermal distribution and solidification rate. The LPBFed Nb521 parts after HIP presented sound tensile properties with the fracture strength of 678.7 ± 1.1 MPa and elongation of $5.91 \pm 0.32\%$. The precipitated nanoscale ZrO_2 underwent plastic deformation after tension, exhibiting the stress-induced tetragonal-to-monoclinic phase transformation and shearing by dislocations. This work presents the LPBFed Nb521 alloy with promising performance by employing low-cost jet-milled HDH powders.

1. Introduction

Due to excellent high-temperature strength, low density in refractory alloys, superior heat and corrosion resistance, niobium (Nb) alloys are promising in usage for high-temperature service mainly in aerospace applications [1–3]. Nb521 (Nb–5W–2Mo–1Zr), a typical dispersion-strengthened Nb alloy, has been widely used in high-temperature sustained propulsion systems [4,5]. Although Nb521 is machinable through conventional forging and machining, it still remains intractable to fabricate sophisticated-geometric and affordable parts to meet the growing demand of lightweight design. Additive manufacturing (AM), however, provides an integrative solution to directly produce Nb521 parts with advantages of complexed geometry, cost saving and high efficiency [6–8], as compared with traditional manufacturing methods.

To date, there have been a few trials on Nb alloys via AM such as Nb521 [9] and C103 (Nb–10HF–1Ti) [6,10,11] in terms of processing,

microstructures, and mechanical properties. For instance, Yang et al. [9] fabricated the Nb521 alloy by electron beam melting (EBM) with focusing on the investigation of (Nb, Zr)C and Nb_2C precipitation behavior. Additionally, Philips et al. [10,11] utilized atomized powders to produce the C103 alloy by both laser powder bed fusion (LPBF) and EBM. The additive manufactured (AMed) C103 alloy exhibited comparable performance to equivalent wrought material [6,10,11]. In addition, Refs. [7,12,13] suggest that the parts made by LPBF show fine geometric features and sound surface finish. Thus, above works demonstrate that LPBF of Nb-based refractory alloys is feasible and effective from the engineering viewpoint in the aerospace field that requires sophisticated geometry and smooth surface finish. One crucial issue remained in the current reports [14–16], however, is that employing spherical powders of Nb-based alloys as the feedstock for AM e.g., LPBF pushes up the budget for final counterparts. To some extent, the high-cost powder feedstock constrains the extensive deployment and future scale-up of Nb-based alloys by AM.

* Corresponding author. Beijing Advanced Innovation Center for Materials Genome Engineering, Institute for Advanced Materials and Technology, University of Science and Technology Beijing, Beijing, 100083, PR China.

** Corresponding author. Beijing Advanced Innovation Center for Materials Genome Engineering, Institute for Advanced Materials and Technology, University of Science and Technology Beijing, Beijing, 100083, PR China.

E-mail addresses: gche098@ustb.edu.cn (G. Chen), quxh@ustb.edu.cn (X. Qu).

<https://doi.org/10.1016/j.msea.2022.143153>

Received 23 January 2022; Received in revised form 10 March 2022; Accepted 12 April 2022

Available online 14 April 2022

0921-5093/© 2022 Elsevier B.V. All rights reserved.

Recently, we developed a novel and facile method *i.e.*, jet milling to modify low-cost hydride-dehydride (HDH) CP-Ti powders that were successfully employed in LPBF [17]. Herein, jet milling is conducted in an inert gas environment in a few minutes, where intensive shape modification takes place due to collision and friction among powders [18]. The resultant particle flowability fulfills the acquirement of AM yet with acceptable impurity control owing to the usage of inert gas. Likewise, the present study used jet milling to produce HDH Nb521 powders that were subsequently employed as the raw feedstock in LPBF. The objective of this work is to investigate microstructures and tensile properties of the LPBF made (LPBFed) Nb521 alloys with a particular focus on the precipitation effect.

2. Experimental

2.1. Powder feedstock

The raw Nb521 alloy powders were house made by the hydrogenation and dehydrogenation process using the pre-alloyed plates purchased from Shaanxi Xingshuo New Material Technology Co., Ltd. P.R. China. Then, jet milling was used to modify powders to improve the sphericity and flowability. Detailed processing information of jet milling can be found in Ref. [17]. Chemical compositions of the jet-milled HDH Nb521 powders are shown in Table 1 with the oxygen level of 0.18 wt%. The mean particle size (D_{50}) of the jet-milled powders is 47.7 μm , determined using a laser diffraction analyzer (Mastersizer 3000, UK). Particle flowability was determined as per the ASTM B213-13 standard using a Hall flowmeter.

2.2. Laser powder bed fusion

The jet-milled Nb521 powders were employed as raw feedstock for LPBF processing (SLM Solutions 125HL, Germany) equipped with a 400 W fiber laser in the continuous-wave mode. The focus diameter of laser is approximately 75 μm , while high-purity argon gas flow was applied during the entire LPBF process. Prior to processing, the Nb521 substrate was preheated to 473 K to reduce residual stress. LPBF processing parameters were chosen with a powder layer thickness of 30 μm , a laser power of 180 W, a scanning speed of 400 mm/s, and a hatch distance of 0.12 mm, respectively. The scanning strategy was chosen as the zigzag pattern, while the scanning direction between each layer was increased by 33° with a limitation window of 90°. In addition, the LPBFed Nb521 parts were subjected to HIP at 1773 K with a pressure of 150 MPa for 2 h to achieve full densification and alleviate the internal stress caused by rapid cooling during LPBF. Here, the nomenclature of HIPed is abbreviated for the LPBFed parts after the HIP treatment.

2.3. Microstructural characterization

Analysis of interstitial (O, N, H) contents of samples was carried out with an inert gas fusion analytical instrument (Leco-Tch600, USA), while carbon (C) content was measured using a carbon and sulfur analyzer (Leco-CS844, USA). Alloying contents (W, Mo, Zr) were measured using an inductively coupled plasma atomic emission spectrometer (Agilent 730 ICP-OES, USA). Phase constituent was examined by X-ray diffraction (XRD, Rigaku Ultima IV, Japan) with Cu K α radiation. Density and porosity of the fabricated parts were determined using the Archimedes method as specified in the ASTM B962 standard. Microstructures of the fabricated parts were observed by scanning electron

microscopy (SEM, JSM-SU8100, Japan). Samples for the SEM observation were etched in etchant (HF: HNO₃: H₂O = 1: 1: 1). Grain orientation of samples was characterized using an SEM (JSM-7900F, Japan) equipped with an electron backscatter diffraction (EBSD) detector (Symmetry, UK). EBSD maps were recorded with a step size of 1.0 μm . Images of high-resolution transmission electron microscopy (HRTEM) and elemental distribution data were determined using a transmission electron microscope (TEM, FEI Tecnai G2 F20, USA) equipped with an energy dispersive spectrometer (EDS) detector (Bruker XFlash 5030, Germany). TEM samples perpendicular to the building direction (BD) were prepared by mechanically grinding to a thickness below 80 μm and then thinning by ion milling (Gatan 691, USA).

2.4. Tension test

The HIPed Nb521 sheets perpendicular to the BD were cut into tensile bars with a gauge length of 10.0 mm via a wire cutter. Tensile properties of the Nb521 bars were determined on an universal testing machine (WDW-10E, P.R. China) using an extensometer with a cross-head speed of 0.4 mm/min at room temperature. Three samples were tested to ensure the value repeatability for each experiment, and average tensile properties were given.

2.5. Simulation of thermal profile in the melt pool

In order to simulate the moving laser energy source and determine the thermal profile associated during LPBF, a proposed model was used to conduct simulations in the ANSYS Fluent code using the Finite Volume Method. The surface heat source of laser $q(x, y, z, t)$ was assumed to exhibit a Gaussian distribution with the following heat transfer equation [19]:

$$q(x, y, z, t) = \frac{2AP}{\pi r_0^2} \exp \left\{ \frac{-2[(x - v_s t)^2 + y^2]}{r_0^2} \right\} \quad (1)$$

where P is the laser power, r_0 is the laser beam radius, v_s is the scanning speed, and A is the laser absorptivity, respectively. The density ρ , specific heat capacity C , thermal conductivity k , surface tension gradient $d\gamma/dT$, laser absorptivity A , liquidus temperature and solidus temperature for the Nb521 alloy were chosen as 8.85 g/cm³, 280 J/kg·K, 48.7 W/m·K, 3×10^{-4} N/m·K, 0.5, 2750 K and 2698 K, respectively. The calculation region was set to $0.4 \times 0.6 \times 1.0$ mm.

Two critical solidification parameters are the solidification rate V (mm/s), and temperature gradient G (K/mm) [20]. The G and V with distance from the bottom to top of the melt pool were calculated based on the following data. The solidification rate V is related to the scanning speed v_s through the inclination angle α between the scanning direction and melt pool boundary *i.e.*, $V = v_s \cdot \cos \alpha$ [21]. The temperature gradient G can be obtained using the formula $\dot{T} = G \times V$, where $|\dot{T}|$ (K/s) is cooling rate calculated by dT/dt at the liquid/solid interface. The relationship between the precipitation behavior and solidification process can be revealed through this simplified model.

3. Results

3.1. Powder characteristics

Fig. 1(a) shows the particle morphology of the jet-milled HDH Nb521 powders, which exhibits grounded and smooth surface appearance. The particle flowability of the jet-milled powders is determined as 41.1 ± 0.3 s/50 g. Fig. 1(b) presents the XRD results of the jet-milled Nb521 powders and LPBFed products, which indicate the Nb521 powders and LPBFed parts both exhibit a typical body centered cubic (BCC) crystal-line structure without significant peak shift.

Table 1
Chemical compositions (wt.%) of the Nb521 samples.

Sample	O	C	H	N	W	Mo	Zr	Nb
Jet-milled	0.18	0.021	0.007	0.023	5.30	1.67	1.60	Bal.
LPBFed	0.25	0.012	0.007	0.025	5.55	1.78	1.63	Bal.

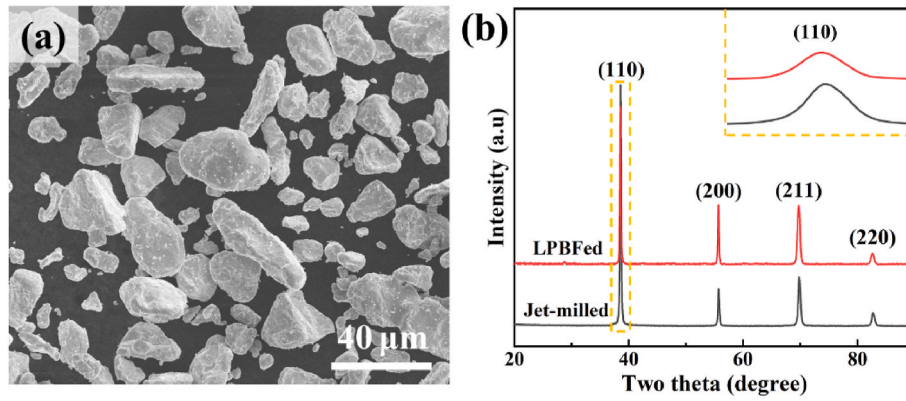


Fig. 1. (a) SEM image of the jet-milled powders, and (b) XRD patterns of the jet-milled powders and LPBFed parts.

3.2. Microstructural characterization

In this work, the optimal relative density of HIPed part reaches $99.2 \pm 0.1\%$. Fig. 2 presents the EBSD analysis of the LPBFed (Fig. 2(a) and (b)) and HIPed (Fig. 2(c) and (d)) samples in terms of crystal orientation map and pole figure. The microstructure for the LPBFed and HIPed samples seems similar with each other. It can be observed from Fig. 2(a) and (c) that there are a few equiaxed grains at the melt pool boundaries for both samples. By contrast, more columnar grains with a larger size are distributed inside the melt pool. The bi-modal microstructure may be related to the thermal profile associated with the cooling rate difference at varying regions in the melt pool during LPBF [22,23]. From the crystal orientation maps parallel to the BD, the columnar crystals grew either along the BD or opposite to the heat flux direction. The pole figures of the LPBFed and HIPed samples perpendicular to the BD in Fig. 2 (b) and (d) do not exhibit preferred texture orientation. However, in the case of parallel to the BD, there is a slight $(100) < 111 >$ texture for the LPBFed sample (Fig. 2(b)), while it is weakened after HIP as proved in Fig. 2(d).

Fig. 3 shows the representative SEM images of the LPBFed and HIPed samples. The grain size along the BD for both samples is roughly in the range of 20–80 μm as indicated in Fig. 3(a) and (c), while the melt pool width ranges from 130–160 μm (Fig. 3(b) and (d)). Further careful observation of Fig. 3(a1) and 3(c1) shows there exist clusters of precipitates dispersed in the Nb solid solution (Nbss) matrix. These quasi-spherical precipitates are with the grain size of ca. 50 nm. It is noted that the precipitate population is different at each side nearby fusion boundaries (FBs) i.e., more precipitates with a larger size formed at the inner bottom of melt pool as compared to its center area as seen in Fig. 3

(b1) and 3(d1). In other words, the precipitation behavior is different in the two particular regions for both samples. Plus, it can be concluded from Fig. 3 that the microstructure is similar for the LPBFed and HIPed samples. To investigate the composition of the nanoscale precipitates, EDS elemental mapping was conducted for the HIPed part as presented in Fig. 3(e), which indicates the existence of oxygen segregation within or nearby the precipitates.

To further confirm the constituent and crystal structure of the nanoscale quasi-spherical precipitates, TEM was carried out to investigate for the HIPed sample. It can be seen from Fig. 4(a) that there are a few quasi-spherical precipitates with the size of ca. 50 nm dispersed in the Nbss matrix, consistent with the findings in Fig. 3(a1) and 3(c1). The corresponding EDS maps indicate the precipitate is consisted of Zr and O elements, as shown in Fig. 4(b). Fig. 4(d) is the interface between the precipitate and Nbss matrix in Fig. 4(c), which exhibits a well-bonded relationship. The fast fourier transform (FFT) analysis in Fig. 4(e) further confirms the nanoscale precipitate is the tetragonal zirconia ($t\text{-ZrO}_2$) phase. It reveals that the (002) crystal plane of $t\text{-ZrO}_2$ is parallel to the (101) crystal plane of Nbss. To further visualize the interface relationship, an inverse FFT image was constructed as shown in Fig. 4(f) based on Fig. 4(d), which presents that the parallel crystal plane spacing of ZrO_2 and Nbss is 0.264 and 0.238 nm, respectively. The lattice mismatch between the Nbss matrix and $t\text{-ZrO}_2$ precipitate is estimated as 10.2%. Therefore, the Nbss/ ZrO_2 interface can be determined as a semi-coherent structure with a few periodically spaced misfit dislocations.

3.3. Tensile properties

Typical engineering tensile stress-strain curves of the HIPed samples

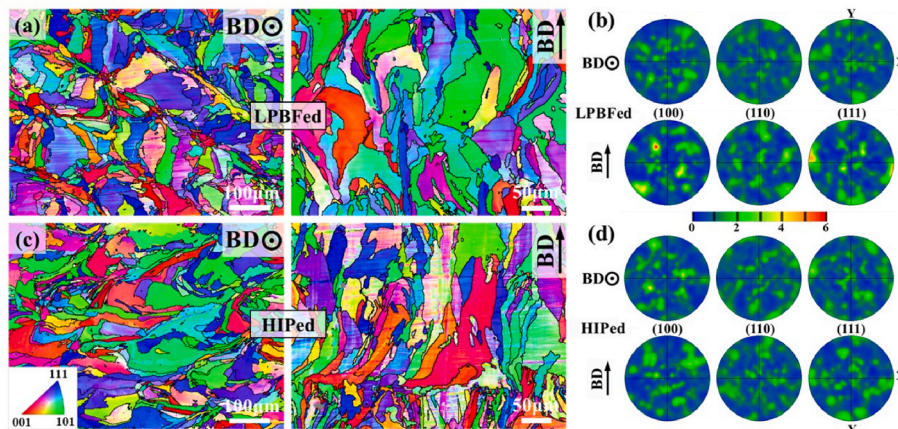


Fig. 2. EBSD analysis of the LPBFed and HIPed samples. (a) Crystal orientation maps, and (b) pole figures of the LPBFed sample; (c) Crystal orientation maps, and (d) pole figures of the HIPed sample. The arrows indicate the building direction (BD).

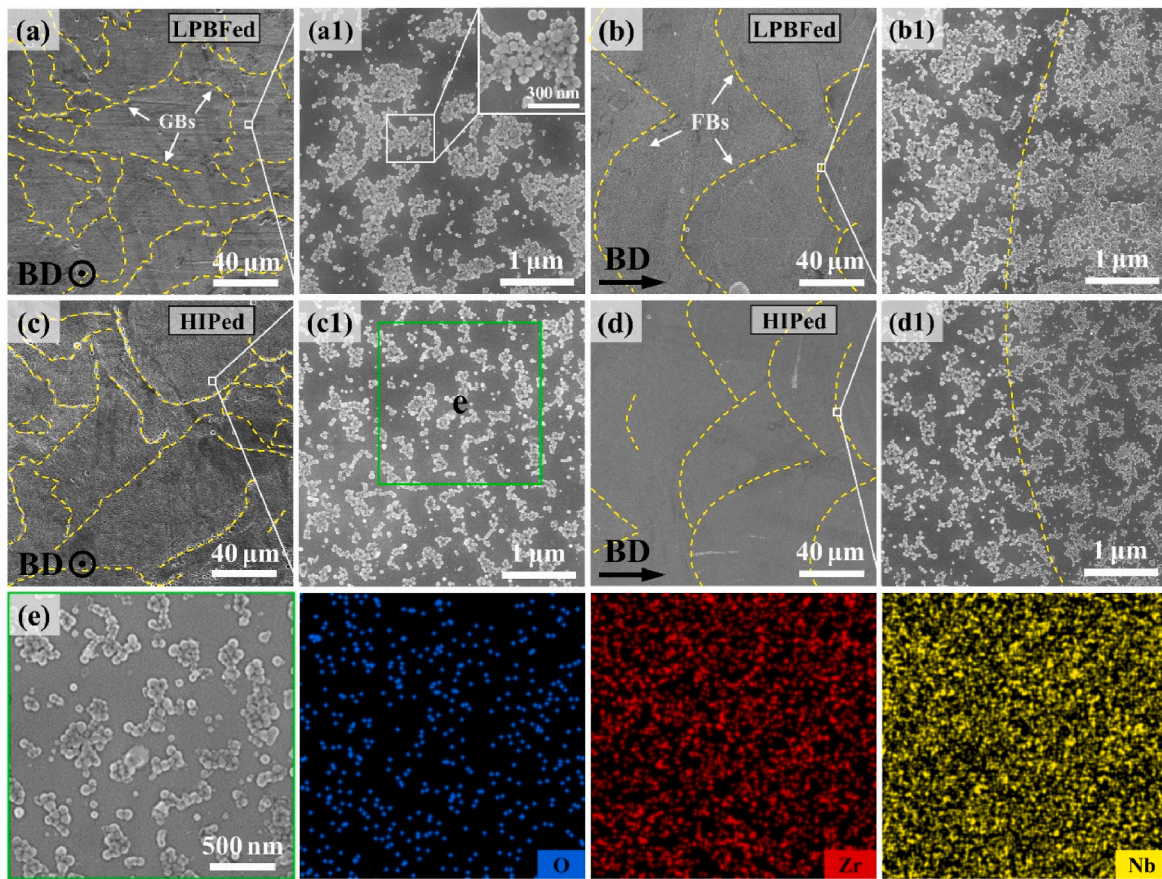


Fig. 3. SEM images of the LPBFed and HIPed samples. The LPBFed sample (a) perpendicular to the BD and (b) parallel to the BD; the HIPed sample (c) perpendicular to the BD and (d) parallel to the BD; (a1), (b1), (c1) and (d1) are enlarged images of the marked region in (a), (b), (c) and (d), respectively. (e) EDS elemental mappings of the marked region in (c1).

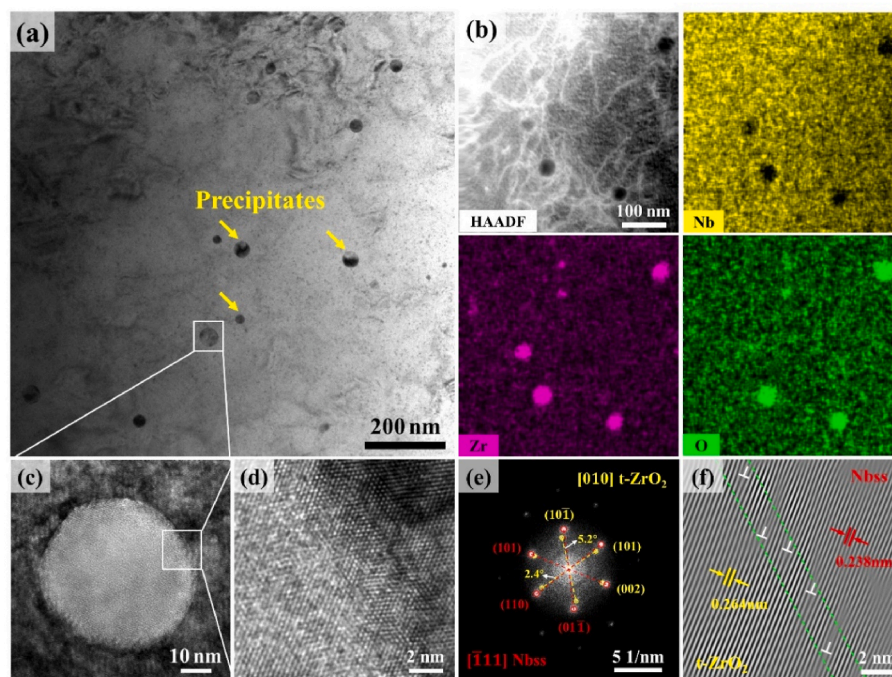


Fig. 4. TEM and HRTEM images of the HIPed sample. (a) Bright field image showing quasi-spherical nano-scale precipitates dispersed in the matrix; (b) HAADF image and corresponding EDS maps for the precipitates; (c) Enlarged HRTEM image of the marked region in (a); (d) Enlarged image of the marked region in (c); (e) FFT image of (d); and (f) corresponding inverse FFT image of (d).

are shown in Fig. 5(a). The fracture tensile strength and elongation of the HIPed samples are 678.7 ± 1.1 MPa and $5.91 \pm 0.32\%$, respectively. The tensile properties in this study together with the reported results of the Nb521 alloys by other routes are summarized and compared in Fig. 5(b). It is noted that our work has achieved superior fracture tensile strength, although its ductility is relatively inferior as compared to the Nb521 alloys by heat treating (HT) of wrought parts, EBM and laser melting deposition (LMD). The strengthening mechanism associated will be discussed in the following section.

4. Discussion

4.1. Precipitation behavior

In the case of Nb521 alloy, alloying elements of W and Mo are infinite solid solution in the Nb matrix [25,26]. It has been reported, however, the active elemental Zr may prefer to potentially react with interstitial elements e.g., O and C to form non-metallic compounds under certain circumstances [27,28]. It has been reported [29] that the Zr could saliently reduce the oxygen solubility via forming ZrO_2 , e.g., an addition of 1 wt% Zr lowering the solubility of oxygen in Nb at 1273 K from 3800 ppm to below 100 ppm. According to the Zr–O binary phase diagram in Fig. 6(a), the solubility limit of oxygen atoms in the α -Zr phase is ca. 32 at.% when the temperature is ~ 2276 K. Herein, assuming that oxygen atoms were all solid dissolved in the Zr matrix lattice, the calculated mole percentage of α_{Zr} : α_O equals to 53%: 47% based on the oxygen level of 0.25 wt% in the LPBFed sample (Table 1), which obviously exceeds the oxygen solubility limit of 32 at.%. Thus, oxygen atoms are preferred to saturate with the elemental Zr to form the oxide phase (e.g., ZrO_2) from the α -Zr(O) solid solution. Associated free energies of possible reactions to generate the precipitates in the Nb521 alloy system are shown in Fig. 6(b) as calculated with Thermo-Calc by equations as given in Ref. [30,31], which indicates the formation of ZrO_2 is thermal dynamically favorable in this study. It is noted that the potential precipitates in the Nb521 alloy commonly include Nb_2C , (Nb, Zr)C and ZrO_2 , in which Nb_2C is a metastable phase while (Nb, Zr)C is a continuous solid solution of carbide precipitate [9,32,30]. The formation of these precipitates is all favorable nearby the melting point of the Nbss matrix based on the free energy results in Fig. 6(b). However, it is hardly to find the Nb_2C and (Nb, Zr)C precipitates except for ZrO_2 in the LPBFed and HIPed samples (Figs. 3 and 4). It is estimated that the cooling rate during LPBF is so high (10^5 – 10^7 K/s) [12] that suppresses the precipitation of Nb_2C and (Nb, Zr)C phases.

We recall the precipitation behavior and specific distribution of ZrO_2 particles as shown in Fig. 3(b1) and 3(d1). The ZrO_2 particles are preferred to precipitate at the bottom of melt pools. To reveal the precipitation mechanism during LPBF, simulation with a single-track scanning model was carried out to depict the thermal profile in a melt pool. Fig. 7(a) shows the temperature distribution in the simulated

single melt pool with the processing parameters of 180 W laser power and 400 mm/s scanning speed. The experimental and simulated melt pools as a function of temperature profile are plotted and compared in Fig. 7(b), which seems well matched. It can be judged from Fig. 7(b) that the ZrO_2 precipitates are enriched along the melt pool boundaries with a core-shell shape, while the core shell thickness is about 18 μm .

Based on the fundamental solidification theory, the temperature gradient G and solidification rate V mainly dominate the localized solidification conditions [33,34]. Consequently, the curves of G and V at the liquidus front with the increase of distance from the bottom of melt pool in Fig. 7(b) were achieved as presented in Fig. 7(c). Generally, solidification starts at the melt pool border where exists a high temperature gradient G [35,36]. The schematic diagram of the solidification process for a single melt pool of the Nb521 alloy based on the microstructure in Fig. 7(b) is illustrated in Fig. 7(d). ZrO_2 possesses a higher melting point (2995 K) than the Nbss matrix (2698 K), thereby generating precipitation preferentially. Then, the Nbss grains started to grow in the opposite direction to the heat flux when the temperature dropped below the Nb521 liquidus [21]. A local non-equilibrium diffusion model (LNDM) for rapid solidification was given in Ref. [37]:

$$D_b^{LNDM}(V/V_{Db}) = \begin{cases} D_b(1 - V^2/V_{Db}^2); & V < V_{Db} \\ 0; & V > V_{Db} \end{cases} \quad (2)$$

where D_b^{LNDM} is the velocity-dependent effective diffusion coefficient in bulk liquid, V is the interface velocity, and V_{Db} is the solute diffusion velocity, respectively. From Fig. 7(c), the solidification rate V gradually increases with distance from the melt pool bottom. Hence, solute atoms of O and Zr formed ZrO_2 precipitates in the early stages via atomic diffusion in the area nearby the melt pool boundaries, as shown in Figs. 3 and 7(d). When V exceeds V_c , the solute O atoms are favorable to dissolved in the Nbss matrix rather than precipitating (Fig. 7(d)). A sudden change of the precipitate population is owing to the transition from diffusion-controlled to fully thermal-controlled growth [37]. The corresponding distance from the melt pool bottom is roughly 18 μm when the critical solidification rate V_c is estimated as 160 mm/s. Continuous increase of scanning speed v_s pushes up the solidification rate V , and thus shortens the diffusion time [38]. Meanwhile, increasing laser power results in the higher peak temperature and considerably longer liquid lifetime of the melt pool [36], which provides sufficient diffusion time. Therefore, the ZrO_2 precipitation was gradually discouraged with the increase of scanning speed or decrease of laser power. It is expected that the precipitate size and its population could be tunable through tailoring the processing parameters in the future works.

4.2. Effect of precipitation on mechanical properties

In this study, there exists a large number of the ZrO_2 precipitate in the Nbss matrix for the LPBFed and HIPed Nb521 alloys as observed in

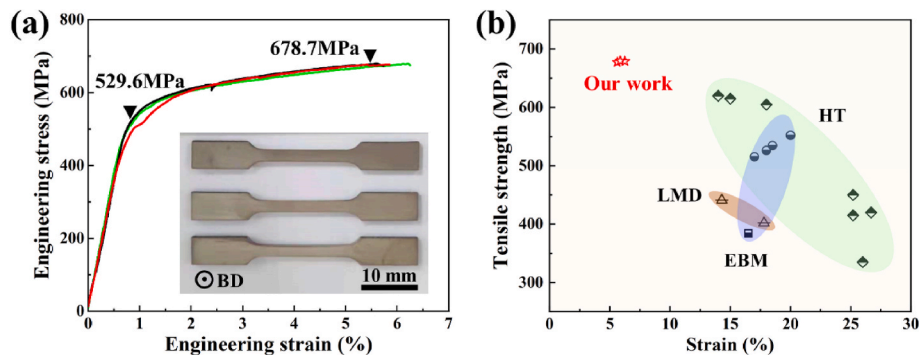


Fig. 5. (a) Engineering tensile stress-strain curves of the HIPed samples. The tensile samples in the inset showing the test direction used in this study. (b) Comparison of tensile properties at fracture between our work and other available literature data [24].

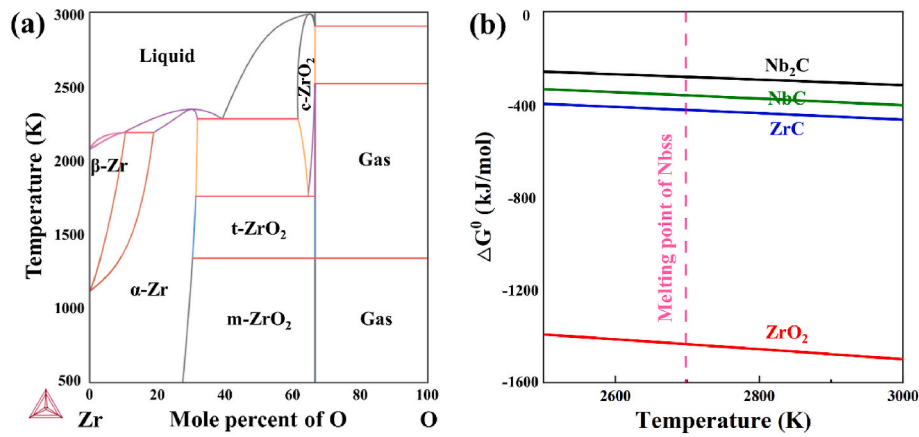


Fig. 6. (a) Zr-O binary phase diagram, and (b) free energy calculations for the reactions to form potential precipitates in this study.

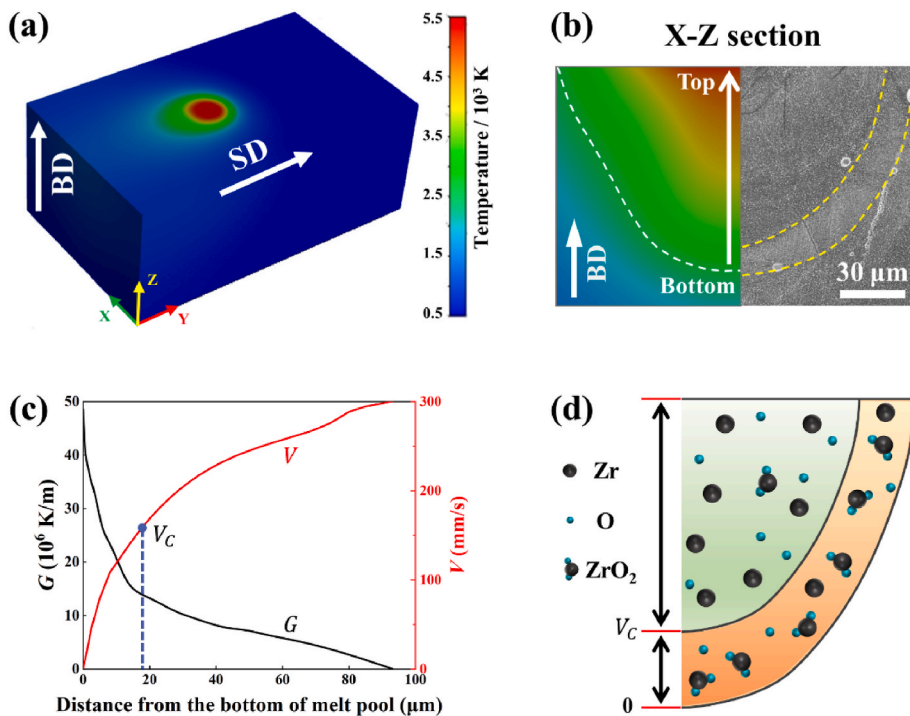


Fig. 7. Thermal profile within a melt pool using a single-track scanning model for the Nb521 alloy during LPBF and schematic illustration of the ZrO₂ precipitation behavior. (a) Thermal distribution of the simulated single melt pool. The white arrows indicate building direction (BD) and scanning direction (SD). (b) Comparison of simulation and experimental results of the melt pool in the X-Z section parallel to the BD. (c) Temperature gradient (G) and solidification rate (V) at the liquidus front from the bottom to top of the melt pool in (b). (d) Schematic diagram of the ZrO₂ precipitation behavior in the single melt pool in this work.

Figs. 3 and 4. Herein, the ZrO₂ precipitates are expected to act as a thermodynamically stable and strengthening phase for the Nb521 alloy [26]. Fig. 8(a) shows the bright-field TEM image of the fractured HIPed Nb521 sample with numerous dislocations remained in the Nbss matrix after tensile deformation. Meanwhile, its dark-field image in Fig. 8(b) indicates a few nanosized ZrO₂ precipitates embedded in the matrix with various sizes and shapes surrounded by dislocations. Fig. 8(c) shows high-density dislocations pile up nearby a typical ZrO₂ particle. This commonly leads to the dislocation pinning effect and effective enhancement of strength due to the Orowan mechanism [39]. Despite of this, the dispersoid strengthening effect usually accompanies with the ductility sacrifice of alloys. With the high-resolution characterization by HRTEM for the nanosized ZrO₂ precipitates as shown in Fig. 8(d) and (e), it can be seen the ZrO₂ particle was severely sheared with generation of defects inside after deformation, as indicated in Fig. 8(d). It is also noted that the (101) plane of the Nbss matrix still remained a semi-coherent relationship with the (002) plane of t-ZrO₂. Another critical point can be found in Fig. 8(e) that the initially stable t-ZrO₂ phase undertook a martensitic transformation to the m-ZrO₂

(monoclinic zirconia) phase with visible plastic deformation. It has been reported that the tetragonal t-ZrO₂ phase could be transformed to the monoclinic m-ZrO₂ phase under certain stresses [40]. In this case, it is believed that the martensitic phase transformation from t-ZrO₂ to m-ZrO₂ is triggered by stress concentration at the neighborhood of the Nbss matrix during deformation, which is termed as the effect of transformation induced plasticity (TRIP) [41] as illustrated in Fig. 8(f). The TRIP effect occurred in our Nb521 alloy could effectively accommodate strains and stress concentration, and thus contribute to the ductilization to some extent [42,43]. In other words, the existence of nanoscale ZrO₂ precipitates made by LPBF plays a crucial role in the strengthening and ductilization for the Nb521 alloy, yielding the promising tensile properties as presented in Fig. 5(b). It needs to point out that the ZrO₂ precipitate size and its population are also important to adjust via tailoring the LPBF processing parameters in future to further optimize the properties.

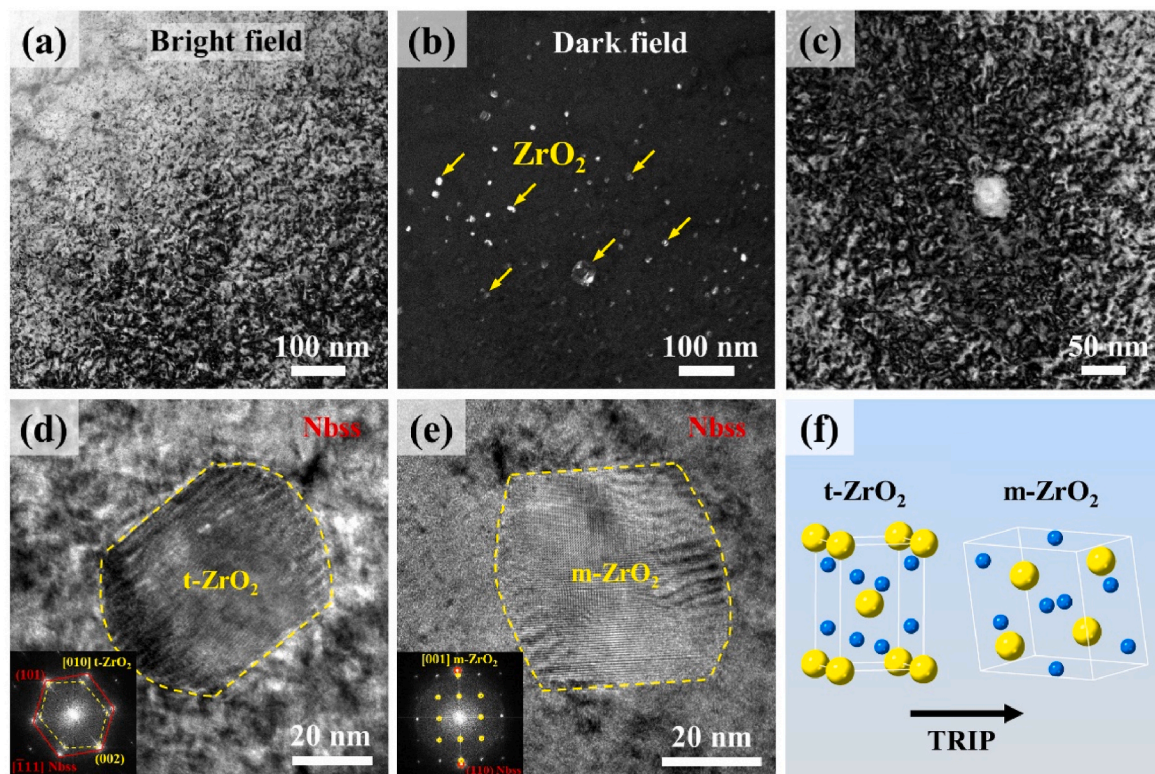


Fig. 8. TEM images showing the microstructures of the HIPed sample after fracture. (a) Bright-field image, and (b) corresponding dark-field image of (a); (c) Typical dislocation pile-up near a ZrO₂ particle. HRTEM images of oxide nanoparticles underwent plastic deformation in (d) and stress-induced phase transformation in (e), the insets of them are corresponding FFT images. (f) Schematic illustrating the unit cell structures of m-ZrO₂ and t-ZrO₂.

5. Conclusions

In summary, we successfully fabricated the Nb521 alloy by LPBF using jet-milled HDH powders. Microstructures of the LPBFed and HIPed samples, precipitation behavior of nanoscale ZrO₂ particles and its effect on the mechanical properties were investigated. The main findings can be summarized as follows:

- (1) The precipitation behavior of nanoscale ZrO₂ particles is investigated by thermodynamics and kinetics associated with the solidification process in terms of temperature gradient and solidification rate within the melt pool during LPBF.
- (2) The ZrO₂ precipitates, exhibiting semi-coherency with the Nbss matrix, enable the dislocation pinning and TRIP effect during tensile deformation, which is important for strengthening and ductilization of the Nb521 alloy.
- (3) The LPBFed sample after HIP achieves promising tensile properties of 678.7 ± 1.1 MPa ultimate strength and $5.91 \pm 0.32\%$ fracture elongation, comparable with others reported in the literature.

CRediT authorship contribution statement

Jianan Chen: Data curation, Writing – original draft, Preparation, Visualization, Investigation. **Wangwang Ding:** Data curation, Writing – original draft, Preparation. **Qiyang Tao:** Data curation, Writing – original draft, Preparation. **Chuanzhen Ma:** Software, Validation. **Cong Zhang:** Software, Validation. **Gang Chen:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Mingli Qin:** Supervision. **Xuanhui Qu:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors appreciate the funds from National Natural Science Foundation of China (No.: 51971036, 52130407, 51774036, 51974029), Shandong Provincial Key Research and Development Program (No.: 2019JZZY010327), and National Key Research and Development Program of China (No.: 2021YFB3701900).

References

- [1] E.N. Sheftel, O.A. Bannykh, Niobium-Base alloys, *Int. J. Refract. Met. H.* 12 (1993) 303–314, [https://doi.org/10.1016/0263-4368\(93\)90038-H](https://doi.org/10.1016/0263-4368(93)90038-H).
- [2] M.S. El-Genk, J.-M. Tournier, A review of refractory metal alloys and mechanically alloyed dispersion strengthened steels for space nuclear power systems, *J. Nucl. Mater.* 340 (2005) 93–112, <https://doi.org/10.1016/j.jnucmat.2004.10.118>.
- [3] B. Li, S. Jiang, K. Zhang, Effects of W content on high temperature oxidation resistance and room temperature mechanical properties of hot-pressing Nb–W alloys, *Mater. Sci. Eng., A* 556 (2012) 15–22, <https://doi.org/10.1016/j.msea.2012.06.053>.
- [4] V.V. Bukhanovskii, Effect of silicide ceramic coatings on the high-temperature strength and ductility of niobium alloy of the Nb–W–Mo–Zr system, *Met. Sci. Heat Treat.* 46 (2004) 78–84, <https://doi.org/10.1023/B:MSAT.0000029607.97403.94>.
- [5] Y. Qiao, K. Zheng, X. Guo, Microstructure formation of silicide coating on Nb–W–Mo–Zr alloy, *Surf. Eng.* 35 (2019) 588–595, <https://doi.org/10.1080/02670844.2018.1547247>.
- [6] N.R. Philips, M. Carl, N.J. Cunningham, New opportunities in refractory alloys, *Metall. Mater. Trans.* 51 (2020) 3299–3310, <https://doi.org/10.1007/s11661-020-05803-3>.
- [7] T. DebRoy, H.L. Wei, J.S. Zuback, T. Mukherjee, J.W. Elmer, J.O. Milewski, A. M. Beese, A. Wilson-Heid, A. De, W. Zhang, Additive manufacturing of metallic

- components – process, structure and properties, *Prog. Mater. Sci.* 92 (2018), <https://doi.org/10.1016/j.pmatsci.2017.10.001>, 112–224.
- [8] T. Zhang, C.-T. Liu, Design of titanium alloys by additive manufacturing: a critical review, *Adv. Powder Mater.* 1 (2021) 100014, <https://doi.org/10.1016/j.apmat.2021.11.001>.
- [9] J. Yang, Y. Huang, B. Liu, C. Guo, J. Sun, Precipitation behavior in a Nb-5W-2Mo-1Zr niobium alloy fabricated by electron beam selective melting, *Mater. Char.* 174 (2021), 111019, <https://doi.org/10.1016/j.matchar.2021.111019>.
- [10] O. Mireles, O. Rodriguez, Y. Gao, N. Philips, Additive manufacture of refractory alloy C103 for propulsion applications, in: *AIAA Propulsion and Energy 2020 Forum*, 2020, pp. 1–13, <https://doi.org/10.2514/6.2020-3500>.
- [11] P.D. Awasthi, P. Agrawal, R.S. Haridas, R.S. Mishra, M.T. Stawowy, S. Ohm, A. Imadoust, Mechanical properties and microstructural characteristics of additively manufactured C103 niobium alloy, *Mater. Sci. Eng., A* 831 (2022), 142183, <https://doi.org/10.1016/j.msea.2021.142183>.
- [12] Z. Liu, D. Zhao, P. Wang, M. Yan, C. Yang, Z. Chen, J. Lu, Z. Lu, Additive manufacturing of metals: microstructure evolution and multistage control, *J. Mater. Sci. Technol.* 100 (2022) 224–236, <https://doi.org/10.1016/j.jmst.2021.06.011>.
- [13] D. Gu, X. Shi, R. Poprawe, D.L. Bourell, R. Setchi, J. Zhu, Material-structure-performance integrated laser-metal additive manufacturing, *Science* 372 (2021), eabg1487, <https://doi.org/10.1126/science.abg1487>.
- [14] O.D. Neikov, S.S. Naboychenko, I.B. Murashova, N.A. Yefimov, Chapter 23 - production of refractory metal powders, in: O.D. Neikov, S.S. Naboychenko, N.A. Yefimov (Eds.), *Handbook of Non-ferrous Metal Powders*, second ed., Elsevier, Oxford, 2019, pp. 685–755.
- [15] K.B. Park, J.M. Park, J. Choi, J.-W. Kang, S.Y. Lee, K. Park, T.-W. Lee, T.-W. Na, H.-K. Park, Preparation of Nb-silicide based alloy powder by hydrogenation-dehydrogenation reaction, *Int. J. Refract. Met. H.* 76 (2018) 180–184, <https://doi.org/10.1016/j.ijrmhm.2018.06.007>.
- [16] D. Zhang, M. Qin, Rafi-ud-din, L. Zhang, X. Qu, Fabrication and characterization of nanocrystalline Nb-W-Mo-Zr alloy powder by ball milling, *Int. J. Refract. Met. H.* 32 (2012) 45–50, <https://doi.org/10.1016/j.ijrmhm.2012.01.008>.
- [17] Q. Tao, Z. Wang, G. Chen, W. Cai, P. Cao, C. Zhang, W. Ding, X. Lu, T. Luo, X. Qu, M. Qin, Selective laser melting of CP-Ti to overcome the low cost and high performance trade-off, *Addit. Manuf.* 34 (2020), 101198, <https://doi.org/10.1016/j.addma.2020.101198>.
- [18] R. Li, M. Qin, Z. Chen, S. Zhao, C. Liu, X. Wang, L. Zhang, J. Ma, X. Qu, Porous structure uniformity investigation of tungsten matrix prepared by jet milled and annealed tungsten powder, *Powder Technol.* 339 (2018) 192–198, <https://doi.org/10.1016/j.powtec.2018.08.021>.
- [19] T.-N. Le, Y.-L. Lo, Effects of sulfur concentration and Marangoni convection on melt-pool formation in transition mode of selective laser melting process, *Mater. Des.* 179 (2019), 107866, <https://doi.org/10.1016/j.matdes.2019.107866>.
- [20] S.A. David, J.M. Vitek, Correlation between solidification parameters and weld microstructures, *Int. Mater. Rev.* 34 (1989) 213–245, <https://doi.org/10.1179/imr.1989.34.1.213>.
- [21] M.-S. Pham, B. Dovggy, P.A. Hooper, C.M. Gourlay, A. Piglion, The role of side-branching in microstructure development in laser powder-bed fusion, *Nat. Commun.* 11 (2020) 749, <https://doi.org/10.1038/s41467-020-14453-3>.
- [22] P. He, R.F. Webster, V. Yakubov, H. Kong, Q. Yang, S. Huang, M. Ferry, J.J. Kruzic, X. Li, Fatigue and dynamic aging behavior of a high strength Al-5024 alloy fabricated by laser powder bed fusion additive manufacturing, *Acta Mater.* 220 (2021), 117312, <https://doi.org/10.1016/j.actamat.2021.117312>.
- [23] O. Andreau, I. Koutiri, P. Peyre, J.-D. Penot, N. Saintier, E. Pessard, T. De Terris, C. Dupuy, T. Baudin, Texture control of 316L parts by modulation of the melt pool morphology in selective laser melting, *J. Mater. Process. Technol.* 264 (2019) 21–31, <https://doi.org/10.1016/j.jmatprotec.2018.08.049>.
- [24] A.V. Ivanchenko, V.P. Belogur, S.I. Karatushin, E.E. Skladnova, K.O. Smirnov, Properties of niobium alloy 5VMTs after deformation and chemical heat treatment, *Met. Sci. Heat Treat.* 39 (1997) 474–476, <https://doi.org/10.1007/BF02469115>.
- [25] R.T. Begley, J.H. Bechtold, Effect of alloying on the mechanical properties of niobium, *J. Less-common Met.* 3 (1961) 1–12, [https://doi.org/10.1016/0022-5088\(61\)90037-6](https://doi.org/10.1016/0022-5088(61)90037-6).
- [26] V.G. Grigorovich, E.N. Sheftel, Physicochemical fundamentals of the development of heat-resistant niobium alloys, *Met. Sci. Heat Treat.* 24 (1982) 472–478, <https://doi.org/10.1007/BF00773152>.
- [27] F. Ostermann, Controlling carbide dispersions in niobium-base alloys, *J. Less-common Met.* 25 (1971) 243–256, [https://doi.org/10.1016/0022-5088\(71\)90148-2](https://doi.org/10.1016/0022-5088(71)90148-2).
- [28] M.G. Ulitchny, R. Gibala, Mechanical properties of interstitial alloys of niobium, *J. Less-common Met.* 30 (1973) 177–180, [https://doi.org/10.1016/0022-5088\(73\)90023-4](https://doi.org/10.1016/0022-5088(73)90023-4).
- [29] R.J. Pérez, A.R. Massih, Thermodynamic evaluation of the Nb-O-Zr system, *J. Nucl. Mater.* 360 (2007) 242–254, <https://doi.org/10.1016/j.jnucmat.2006.10.008>.
- [30] Y. Zeng, P. Zhou, Y. Du, Thermodynamic assessment of the C-Zr-Nb ternary system, *Calphad* 61 (2018) 98–104, <https://doi.org/10.1016/j.calphad.2018.02.009>.
- [31] S.Y. Kwon, I.-H. Jung, Thermodynamic assessment of the Al_2O_3 -ZrO₂, CaO-Al₂O₃-ZrO₂, and Al₂O₃-SiO₂-ZrO₂ systems, *Ceram. Int.* 48 (2022) 5413–5427, <https://doi.org/10.1016/j.ceramint.2021.11.085>.
- [32] B. Vishwanadh, K. Vaibhav, S.K. Jha, K.V. Mirji, I. Samajdar, D. Srivastava, R. Tewari, N. Saibaba, G.K. Dey, Development of Nb-1%Zr-0.1%Co alloy as structural components for high temperature reactors, *J. Nucl. Mater.* 427 (2012) 350–358, <https://doi.org/10.1016/j.jnucmat.2012.04.014>.
- [33] A. Prasad, L. Yuan, P. Lee, M. Patel, D. Qiu, M. Easton, D. StJohn, Towards understanding grain nucleation under additive manufacturing solidification conditions, *Acta Mater.* 195 (2020) 392–403, <https://doi.org/10.1016/j.actamat.2020.05.012>.
- [34] M. Bermingham, D. StJohn, M. Easton, L. Yuan, M. Dargusch, Revealing the mechanisms of grain nucleation and formation during additive manufacturing, *JOM* 72 (2020) 1065–1073, <https://doi.org/10.1007/s11837-020-04019-5>.
- [35] S. Marola, D. Manfredi, G. Fiore, M.G. Poletti, M. Lombardi, P. Fino, L. Battezzati, A comparison of selective laser melting with bulk rapid solidification of AlSi10Mg alloy, *J. Alloys Compd.* 742 (2018) 271–279, <https://doi.org/10.1016/j.jallcom.2018.01.309>.
- [36] Y. Li, D. Gu, Thermal behavior during selective laser melting of commercially pure titanium powder: numerical simulation and experimental study, *Addit. Manuf.* 1–4 (2014) 99–109, <https://doi.org/10.1016/j.addma.2014.09.001>.
- [37] S.L. Sobolev, Local non-equilibrium diffusion model for solute trapping during rapid solidification, *Acta Mater.* 60 (2012) 2711–2718, <https://doi.org/10.1016/j.actamat.2012.01.036>.
- [38] W. Ou, T. Mukherjee, G.L. Knapp, Y. Wei, T. DebRoy, Fusion zone geometries, cooling rates and solidification parameters during wire arc additive manufacturing, *Int. J. Heat Mass Tran.* 127 (2018) 1084–1094, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.08.111>.
- [39] T. Gladman, Precipitation hardening in metals, *Mater. Sci. Technol.* 15 (1999) 30–36, <https://doi.org/10.1179/026708399773002782>.
- [40] J. Wang, R. Stevens, Zirconia-toughened alumina (ZTA) ceramics, *J. Mater. Sci.* 24 (1989) 3421–3440, <https://doi.org/10.1007/BF02385721>.
- [41] J. Su, D. Raabe, Z. Li, Hierarchical microstructure design to tune the mechanical behavior of an interstitial TRIP-TWIP high-entropy alloy, *Acta Mater.* 163 (2019) 40–54, <https://doi.org/10.1016/j.actamat.2018.10.017>.
- [42] Z. Zhang, W. Pantleon, Response of oxide nanoparticles in an oxide dispersion strengthened steel to dynamic plastic deformation, *Acta Mater.* 149 (2018) 235–247, <https://doi.org/10.1016/j.actamat.2018.02.042>.
- [43] D. Huang, Y. Zhuang, Break the strength-ductility trade-off in a transformation-induced plasticity high-entropy alloy reinforced with precipitation strengthening, *J. Mater. Sci. Technol.* 108 (2022) 125–132, <https://doi.org/10.1016/j.jmst.2021.08.061>.